

Sardor Sobirov

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EDUCATION

University TBA

Master of Science in Statistics and Machine Learning

Expected Dec 2028

University of California San Diego (UCSD)

Bachelor of Science in Data Science

La Jolla, CA

Jun 2026

- **Relevant Coursework:** Machine Learning Algorithms, Deep Learning and Neural Networks, Probabilistic AI and Graphical Models, Data Mining and Statistical Learning, Distributed Data Systems and Scalable Analytics.

SKILLS

Languages and ML: Python, SQL, JavaScript/TypeScript, pandas, NumPy, scikit-learn, PyTorch, LightGBM, Hugging Face Transformers.

Data Engineering: PostgreSQL, BigQuery, API ingestion, ETL/ELT, JSON/JSONL, Parquet, NetCDF, data validation, reproducible pipelines.

Analytics and Visualization: A/B testing, hypothesis testing, KPI design, Power BI, Tableau, React/Next.js, Streamlit, D3.js, MapLibre.

EXPERIENCE

Geohazard Data Science Intern | U.S. Department of the Interior (National Park Service)

Mount Rainier National Park, WA

Sep 2026 – Jun 2027

- Develop geospatial digital twins of Mount Rainier river and glacier systems from LiDAR, satellite imagery, and field sensors using Python and ArcGIS Pro to monitor geomorphic change.
- Build statistical and machine-learning workflows for seismic, RFID/BLE tracer, and streamflow data to quantify sediment transport and detect geohazard signals for the NPS Imminent Threats Program.

Data Science Research and Development Intern | Cotality

San Diego, CA

Sep 2025 – Mar 2026

- Built a React/TypeScript data-quality dashboard with MapLibre GL and D3.js, turning notebook-generated CSV/JSON artifacts into reviewer-facing views for Cotality's wildfire-risk anomaly investigation.
- Processed 2.4M+ property records across 58 California counties with BigQuery GIS, SQL, H3 indexing, spatial joins, buffers, and centroid geometry checks to produce QA/QC tables for model review.

Data Engineer Intern | Apptics AI

Walnut Creek, CA

Jun 2025 – Sep 2025

- Built a reproducible Python ETL pipeline ingesting daily paid media performance data from the Meta Ads API into PostgreSQL for 5 DTC brands, consolidating \$1K-\$10K/day per brand spend into a single source of truth for reporting.
- Designed campaign, ad set, and ad-level schemas; standardized 25+ SQL KPIs and validation checks so report issues could be traced from source API pulls through Power BI outputs.

Research and Development Intern | United States Department of Defense

La Jolla, CA

Sep 2024 – Dec 2024

- Conducted 50+ stakeholder interviews using Lean LaunchPad methodology to translate ambiguous user needs into a prioritized product roadmap, ensuring research and development investments were tied to concrete operational use cases.
- Designed a portfolio analytics dashboard in Excel and Power BI mapping technology maturity signals to DoD use cases; presented findings to cross-functional leadership to inform multi-million dollar transition decisions.

PROJECTS

GeoCrop Spatiotemporal Modeling | [Dashboard](#) | [Paper](#) | [GitHub](#)

- Built an end-to-end ML/data pipeline for the 13-state Corn Belt, integrating USDA CDL, MODIS NDVI, and NASA SMAP data into NetCDF cubes, Parquet tables, metadata sidecars, and dashboard-ready artifacts.
- Modeled crop phenology, rotation, soil-moisture anomalies, and crop prediction using HSGP, Dirichlet-Multinomial, NIG Bayesian scoring, and LightGBM; achieved 79.2% accuracy with SHAP and ablation diagnostics.

Wildfire Property Intelligence | [Dashboard](#) | [Paper](#) | [GitHub](#)

- Created a statistical anomaly-review workflow using Empirical Bayes, neighbor divergence, classifier two-sample tests, Moran's I, and static JSON exports for a React/TypeScript dashboard.
- Converted sparse-data artifacts, land-cover differences, and inconsistent category labels into reviewable data-quality signals for wildfire property-risk modeling.